Exp 9

```
clc;
clear;
close all;

% Continuous-Time LTI System: dy(t)/dt + 2y(t) = x(t)
% Assume zero initial conditions: H(s) = Y(s)/X(s) = 1/(s + 2)

num = [1];
den = [1 2];
sys = tf(num, den);

disp('Transfer function:');
sys

disp('Pole:');
disp(pole(sys));

disp('Zeros:');
disp(zero(sys));

t = 0:0.01:5;

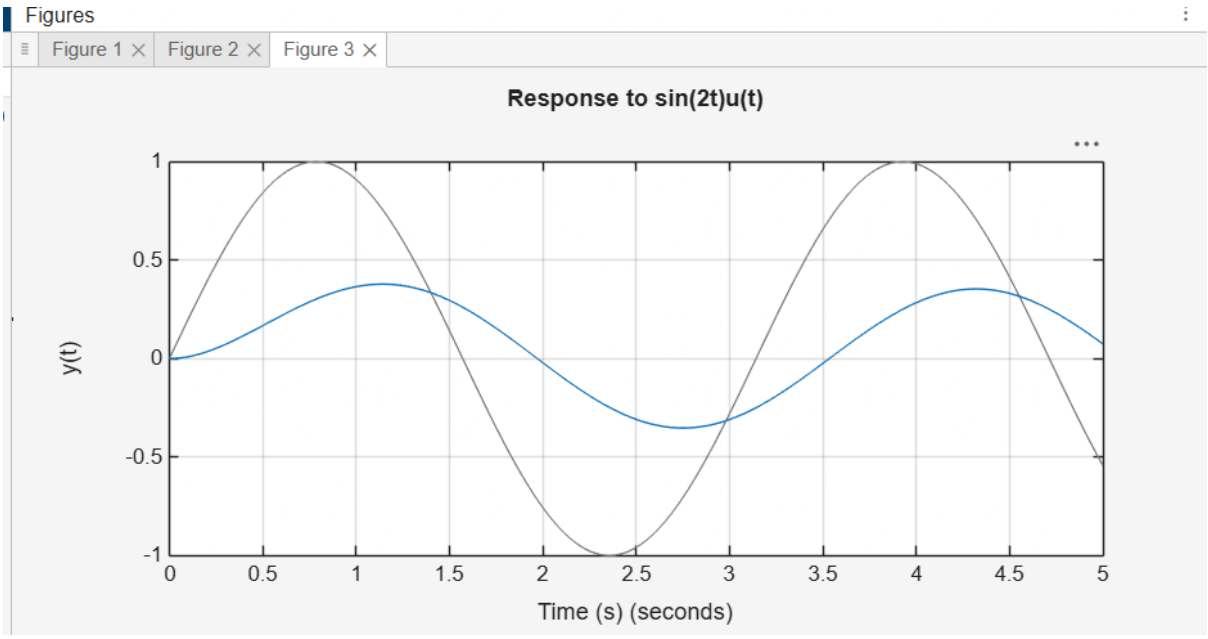
% Impulse Response
figure;
impulse(sys, t);
grid on;
title('Impulse Response');
xlabel('Time (s)');
ylabel('h(t)');

% Step Response
```

```
figure;
step(sys, t);
grid on;
title('Step Response');
xlabel('Time (s)');
ylabel('y(t)');

% Response to x(t) = sin(2t)u(t)
x = sin(2*t);

figure;
lsim(sys, x, t);
grid on;
title('Response to sin(2t)u(t)');
xlabel('Time (s)');
ylabel('y(t)');
```



Exp 10

Exp clc;

```
clear;
close all;

% Experiment 10: Zero-input, zero-state, impulse response and convolution
% System: dy(t)/dt + 2y(t) = x(t)

t = 0:0.01:5;
dt = t(2)-t(1);

num = [1];
den = [1 2];

disp('Transfer function: H(s) = 1/(s+2)');
disp('Pole:');
disp(-2);

% Impulse response
h = exp(-2*t);

figure;
plot(t,h,'LineWidth',1.5);
grid on;
title('Impulse Response h(t)');
xlabel('Time (s)');
ylabel('h(t)');

% Zero-input response: y(0-) = 1
y_zi = exp(-2*t);

figure;
plot(t,y_zi,'LineWidth',1.5);
grid on;
```

```
title('Zero-Input Response');
xlabel('Time (s)');
ylabel('y_{zi}(t)');

% Unit-step input
x = ones(size(t));

% Zero-state response
y_zs = 0.5*(1-exp(-2*t));

figure;
plot(t,x,'--',t,y_zs,'LineWidth',1.5);
grid on;
legend('Input x(t)','Output y_{zs}(t)','Location','best');
title('Unit-Step Input and Zero-State Output');
xlabel('Time (s)');
ylabel('Amplitude');

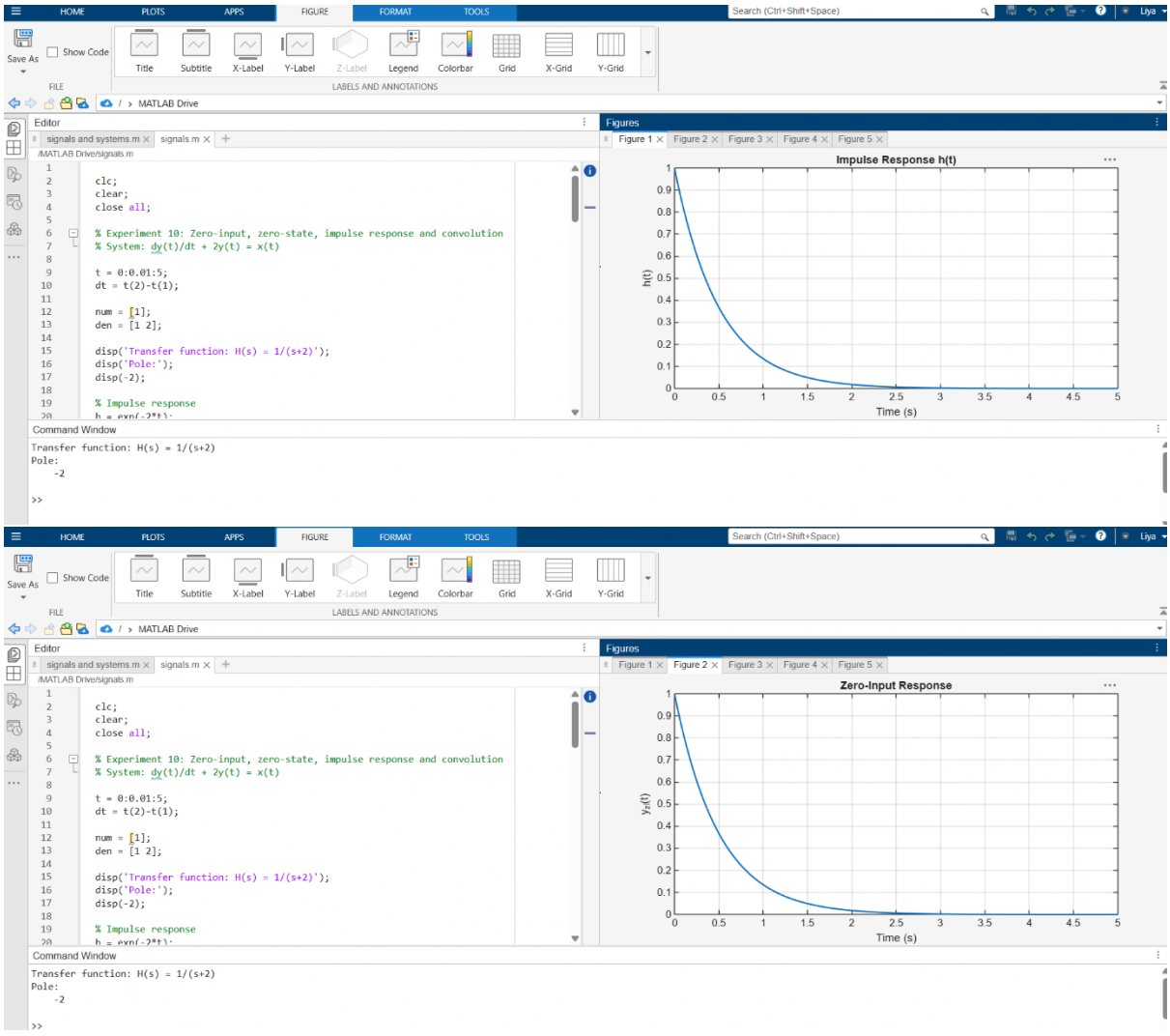
% Numerical convolution
y_conv_full = conv(x,h)*dt;
y_conv = y_conv_full(1:length(t));
y_conv = y_conv - 0.5*dt*(x(1)*h + x*h(1));

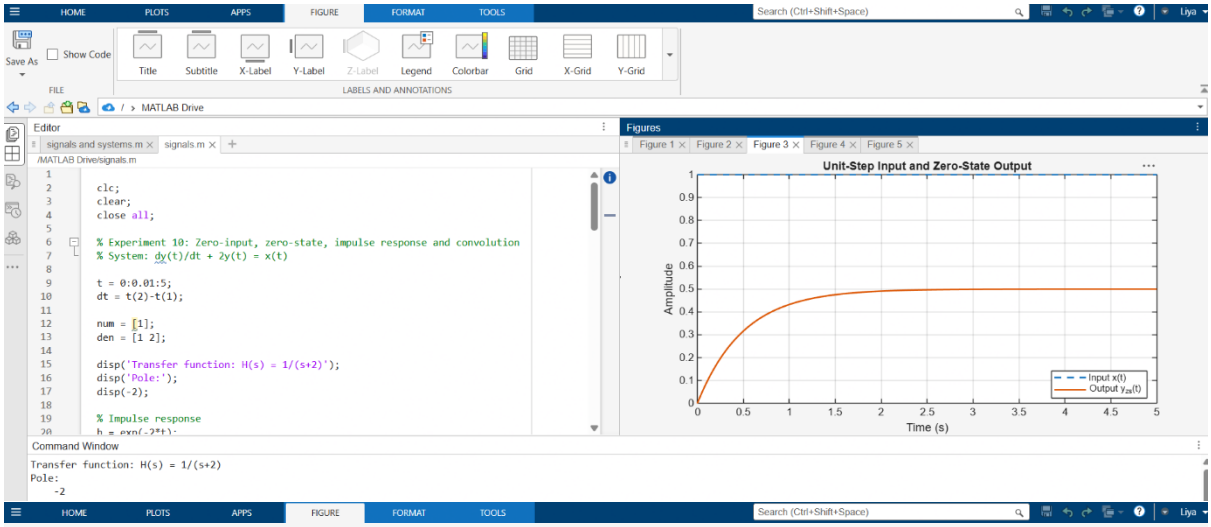
% Total response
y_total = y_zi + y_zs;

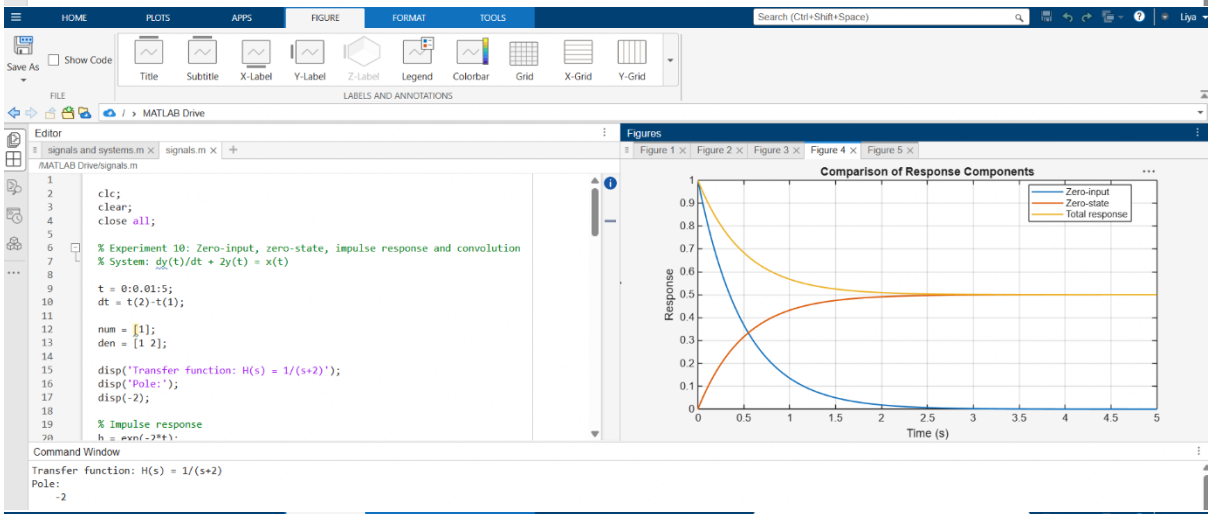
figure;
plot(t,y_zi,t,y_zs,t,y_total,'LineWidth',1.3);
grid on;
legend('Zero-input','Zero-state','Total response','Location','best');
title('Comparison of Response Components');
xlabel('Time (s)');
```

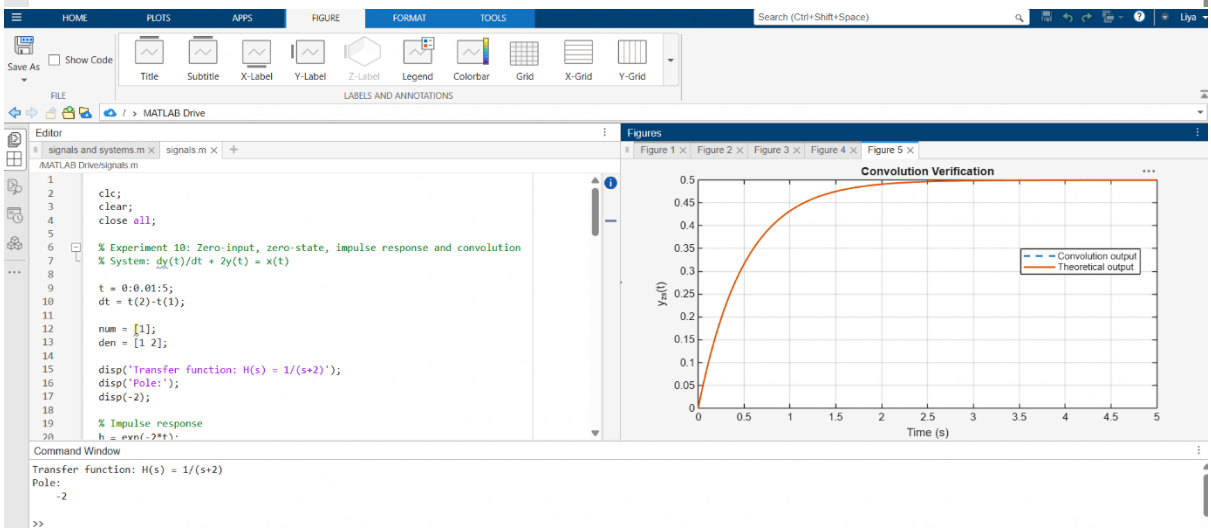
```
ylabel('Response');


% Convolution verification
figure;
plot(t,y_conv,'--',t,y_zs,'-','LineWidth',1.5);
grid on;
legend('Convolution output','Theoretical output','Location','best');
title('Convolution Verification');
xlabel('Time (s)');
ylabel('y_{zs}(t)');
```




HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Liya
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Editor
signals and systems.m
signals.m
/MATLAB Drive/signals.m
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close all;
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% System: dy(t)/dt + 2y(t) = x(t)
t = 0:0.01:5;
dt = t(2)-t(1);
num = [1];
den = [1 2];
disp('Transfer function: H(s) = 1/(s+2)');
disp('Pole:');
disp(-2);
% Impulse response
Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Unit-Step Input and Zero-State Output
Amplitude
Time (s)
Input x(t)
Output y_zs(t)
Command Window
Transfer function: H(s) = 1/(s+2)
Pole:
-2


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Liya
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
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Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Comparison of Response Components
Response
Time (s)
Zero-input
Zero-state
Total response
Command Window
Transfer function: H(s) = 1/(s+2)
Pole:
-2


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Liya
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
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Grid
X-Grid
Y-Grid
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LABELS AND ANNOTATIONS
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disp('Pole:');
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Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Convolution Verification
y_zs(t)
Time (s)
Convolution output
Theoretical output
Command Window
Transfer function: H(s) = 1/(s+2)
Pole:
-2
>>